

Sentiment Analysis Case Study Rubric

DS 4002 – Spring 2026 – Austin Blackburn

Due: May 8

Submission format: Upload link to GitHub repository on Canvas

Individual Assignment

Why am I doing this? Every day, millions of users on Letterboxd express their opinions about films through written reviews and star ratings. But how well does the language in a review actually reflect the rating attached to it? Pretrained sentiment models are powerful tools used across industry to classify opinion at scale, but their accuracy depends heavily on the nature of the text they are given. This case study gives you the opportunity to stress-test one of these models against real, informal, user-generated text and investigate where and why it succeeds or fails.

What am I going to do? The GitHub repository for this case study can be found at <https://github.com/AustinB2319/CS3-DS4002>. Begin by reading the Hook Document and the articles included in the MATERIALS folder to familiarize yourself with the project context. Then navigate to the DATA folder and access the Letterboxd Top 250 reviews dataset. Spend time exploring the data to understand its structure before beginning your analysis.

Your task is to apply a pretrained HuggingFace transformer-based sentiment model to the review text in the dataset and evaluate how well its predictions align with each reviewer's actual star rating. You will convert star ratings into sentiment categories (negative, neutral, positive) to serve as ground truth labels. You will then engineer features from the review text, specifically review length, emoji usage, capitalization patterns, and language, and analyze how each of these characteristics affects model accuracy. Finally, you will write a one-page reflection on your process and findings.

Your final deliverables should include:

- GitHub repository
- Single page PDF reflection

All of this will be submitted electronically via a link to your GitHub repository on Canvas.

Tips for success:

- Comment your code thoroughly so your process is easy to follow
- Think carefully about how you map star ratings to sentiment categories as your choices here will affect your results
- Accuracy alone can be misleading on imbalanced datasets; consider additional metrics that tell a fuller story
- Keep all materials, code, data, and outputs inside the GitHub repository

How will I know I have Succeeded? You will meet expectations on this Case Study when you follow the criteria in the rubric below.

Formatting	• One GitHub repository submitted via link on Canvas. • The repository should contain: ○ a Data folder with the Letterboxd dataset ○ a Scripts and Output folder with all code, generated plots, and CSVs ○ a single page PDF reflection
Data	• Goal: All data used in the analysis is contained in the Data folder of the repository. • The original raw dataset (LetterboxdTop250-5000reviews.csv) must be included. If the file is too large for GitHub, include a link to the original Kaggle source in the README. • The cleaned, review-level dataset produced by your cleaning script must also be included. • Any additional datasets used beyond what is provided in the repository must be added to the DATA folder as well.
Scripts and Output	• Goal: All code is clean, commented, and contained in the Scripts and Output folder of the repository. • Your code must include: ○ (1) a data cleaning script that reshapes the raw dataset so each row represents one review and one rating, removes missing values, and engineers the following features: review length, emoji count, capitalization ratio, and language detection. ○ (2) an analysis script that loads the cleaned dataset, applies a pretrained HuggingFace transformer model to predict sentiment, maps star ratings to sentiment categories as ground truth, applies a neutral confidence threshold to handle ambiguous predictions, computes at least three evaluation metrics (e.g. accuracy, Macro F1, confusion matrix), and generates accuracy-by-feature plots for all four engineered features
Reflection	• Goal: Reflect on your results, process, and learning. • Analyze your results and main takeaways (2 paragraphs). • Discuss challenges you encountered and how you addressed them (1 paragraph). • Reflect on what you learned and what you would do differently or incorporate into future work (1 paragraph). • PDF format • One page maximum

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